

Abstract

Standard quantum mechanics textbooks almost exclusively treat tunneling through rectangular barriers, since the constant-potential Schrödinger equation is one of the few cases admitting a closed-form solution. This paper asks a straightforward question: how much does that simplification actually matter? To find out, we built a one-dimensional Schrödinger equation solver from scratch in pure Python, validated it to sub-0.04% accuracy against analytical models, and used it to compare tunneling through five distinct barrier profiles (rectangular, Gaussian, triangular, parabolic, and exponential) under identical height and width constraints. We find that the choice of geometry changes the transmission probability by several orders of magnitude — at a barrier width of $W = 5.0$, the exponential profile transmits roughly 2.3×10^6 times more than the rectangular baseline. We define a Shape Factor S from the WKB integral and show that it tracks these numerical differences quantitatively. As secondary applications, the solver is used to reproduce double-well tunnel splitting, anharmonic Morse energy levels, periodic band structures, Anderson localization, resonant tunneling, and time-dependent wave packet dynamics.

Numerical Investigation of Quantum Tunneling Through Non-Standard Barrier Geometries

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1 Introduction

The time-independent Schrödinger equation (TISE) can be solved analytically for only a handful of idealized potentials: the infinite square well, the harmonic oscillator, the Coulomb potential, and a few others. In courses on introductory quantum mechanics, tunneling is therefore taught almost exclusively with rectangular barriers, where $V(x)$ is piecewise constant and the wavefunction is a simple exponential decay inside the classically forbidden region.

Real physical barriers, however, are never rectangular. The Coulomb barrier in nuclear fusion is roughly parabolic. The image-charge-corrected work function in field emission is triangular (the Fowler–Nordheim model). Molecular potential energy surfaces are smooth curves. If the shape of the barrier matters for the tunneling rate, then every classroom calculation based on the rectangle is, at best, an order-of-magnitude estimate.

The goal of this study is to quantify *how much* barrier geometry matters. We compare five profiles under controlled conditions and introduce a dimensionless Shape Factor S derived from the WKB approximation to explain the observed differences. As a secondary goal, we validate our numerical solver against a variety of well-understood quantum systems to demonstrate that the same codebase can be applied broadly.

2 Computational Methodology

2.1 Finite Difference Discretization

The 1D TISE is discretized on a uniform grid of N points with spacing Δx . Working in natural units ($\hbar = m = 1$), the central-difference approximation

to the kinetic energy operator yields a tridiagonal Hamiltonian matrix:

$$H_{ii} = \frac{1}{(\Delta x)^2} + V_i, \quad H_{i,i\pm 1} = -\frac{1}{2(\Delta x)^2}. \quad (1)$$

The eigenvalue problem $H\psi = E\psi$ is then solved numerically.

2.2 Eigenvalue Solver

Eigenvalues are isolated by bisection on the Sturm sequence count of the tridiagonal matrix. For a given trial energy μ , the number of eigenvalues below μ can be determined exactly by a single $O(N)$ recurrence, without forming any large intermediate matrices. The search bounds are set using the Gershgorin circle theorem to guarantee that no eigenvalue is missed. Bisection proceeds to a tolerance of 10^{-12} .

Once an eigenvalue is found, the corresponding eigenvector is obtained by inverse iteration with partial pivoting, followed by Gram–Schmidt orthogonalization against all previously found states and normalization to $\int |\psi|^2 dx = 1$.

The entire solver is implemented in pure Python (approximately 350 lines) with no external dependencies.

2.3 Transmission Coefficient

For scattering problems, we use a backward-propagation transfer matrix method. Starting from a pure outgoing plane wave $\psi(x) = e^{ikx}$ on the far right, the finite-difference recurrence is iterated leftward through the potential. The incident and reflected amplitudes are then extracted from the asymptotic wavefunction on the left, and the transmission coefficient is $T = 1/|A_{\text{inc}}|^2$.

2.4 Solver Verification

Table 1 compares the first six numerical eigenvalues of the infinite square well and the harmonic oscillator against their known analytical values. All errors are below 0.04%.

2.5 Convergence with Grid Refinement

To confirm that the errors in Table 1 are controlled by the grid spacing and not by a systematic bug, we computed the ground-state energy of the infinite square well at six different resolutions. Table 2 and Figure 1 show that the error decreases quadratically with Δx , verifying the expected $O(\Delta x^2)$ second-order accuracy of the central-difference operator.

Table 1: Numerical vs. analytical eigenvalues. Infinite well: $L = 1.0$, $N = 300$ interior grid points. Harmonic oscillator: $\omega = 1.0$, $N = 500$, $x \in [-8, 8]$.

System	State	Analytical	Numerical	Error (%)
Infinite Well	$n = 1$	4.9348	4.9348	0.0009
Infinite Well	$n = 2$	19.7392	19.7385	0.0036
Infinite Well	$n = 3$	44.4132	44.4096	0.0082
Infinite Well	$n = 4$	78.9568	78.9454	0.0145
Infinite Well	$n = 5$	123.370	123.342	0.0227
Infinite Well	$n = 6$	177.653	177.595	0.0327
Harmonic Osc.	$n = 0$	0.5000	0.4999	0.0064
Harmonic Osc.	$n = 1$	1.5000	1.4998	0.0107
Harmonic Osc.	$n = 2$	2.5000	2.4996	0.0167
Harmonic Osc.	$n = 3$	3.5000	3.4992	0.0230
Harmonic Osc.	$n = 4$	4.5000	4.4987	0.0293
Harmonic Osc.	$n = 5$	5.5000	5.4980	0.0356

Table 2: Convergence of the ground-state energy of the infinite square well ($E_{\text{exact}} = \pi^2/2 \approx 4.93480$). Error scales as $O(\Delta x^2)$.

N	Δx	$E_{\text{numerical}}$	Error (%)
50	0.0196	4.93324	0.0316
100	0.0099	4.93440	0.0081
200	0.0050	4.93470	0.0020
400	0.0025	4.93478	0.0005
800	0.0012	4.93480	0.0001
1600	0.0006	4.93480	0.00003

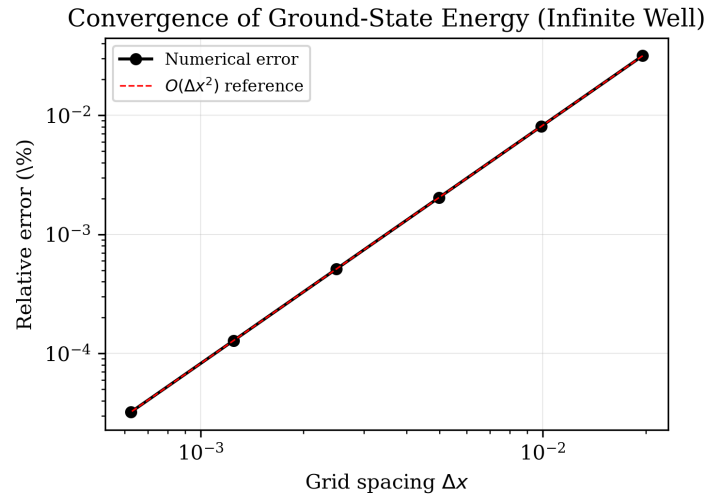


Figure 1: Log-log convergence plot of relative error versus grid spacing Δx for the infinite well ground state, confirming $O(\Delta x^2)$ theoretical scaling.

3 Results: Barrier Geometry and Tunneling

3.1 Experimental Setup

All tunneling calculations use a grid of $N = 800$ points on $x \in [-10, 10]$ ($\Delta x = 0.025$). The barrier maximum is $V_0 = 5.0$ and the incident particle energy is $E = 1.0$ (20% of V_0). The barrier width W is swept from 0.5 to 5.0. Five profiles are compared, all sharing the same peak height V_0 and the same support width W :

- **Rectangular:** $V(x) = V_0$ for $|x| \leq W/2$.
- **Parabolic:** $V(x) = V_0 [1 - (2x/W)^2]$ for $|x| \leq W/2$.
- **Triangular:** $V(x) = V_0 [1 - |2x/W|]$ for $|x| \leq W/2$.
- **Gaussian:** $V(x) = V_0 \exp(-x^2/2\sigma^2)$, $\sigma = W/4$.
- **Exponential:** $V(x) = V_0 \exp(-|4x/W|)$ for $|x| \leq W/2$.

Figure 2 illustrates the five potential profiles superimposed at $W = 3.0$, highlighting the stark difference in enclosed potential area.

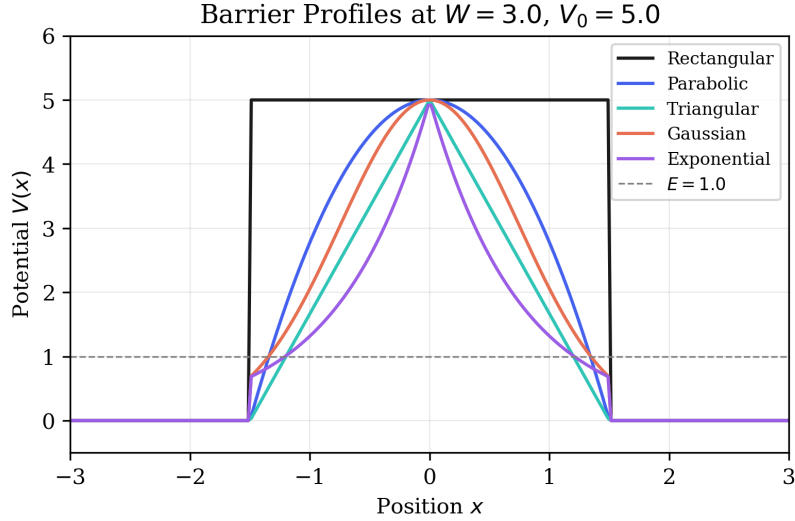


Figure 2: Overlay of the five barrier geometries evaluated under fixed $W = 3.0$ and $V_0 = 5.0$, with incident energy $E = 1.0$ indicated.

It is important to note that while all profiles share the same peak height, their integrated areas (the “amount of potential” the particle must traverse) differ substantially. The rectangular barrier has the largest area, and the exponential has the smallest. This difference in area is, in fact, the key physical reason for the different transmission rates.

3.2 Transmission Results

Table 3 shows the computed transmission coefficient T for each barrier profile at five representative widths, and Figure 3 shows the continuous logarithmic decay across the entire parameter sweep.

Table 3: Transmission coefficient T for each barrier profile at selected widths. Parameters: $V_0 = 5.0$, $E = 1.0$.

W	Rectangular	Parabolic	Triangular	Gaussian	Exponential
1.0	8.9×10^{-3}	5.2×10^{-2}	1.3×10^{-1}	7.5×10^{-2}	1.8×10^{-1}
2.0	3.1×10^{-5}	9.7×10^{-4}	5.7×10^{-3}	1.9×10^{-3}	1.2×10^{-2}
3.0	1.1×10^{-7}	1.7×10^{-5}	2.4×10^{-4}	4.8×10^{-5}	7.2×10^{-4}
4.0	3.7×10^{-10}	3.0×10^{-7}	1.0×10^{-5}	1.2×10^{-6}	4.6×10^{-5}
5.0	1.3×10^{-12}	5.2×10^{-9}	4.4×10^{-7}	3.0×10^{-8}	3.0×10^{-6}

At $W = 5.0$, the exponential barrier transmits 3.0×10^{-6} compared to 1.3×10^{-12} for the rectangular barrier — a ratio of approximately 2.3×10^6 . The discrepancy grows with increasing width because the exponents in the tunneling decay are proportional to W .

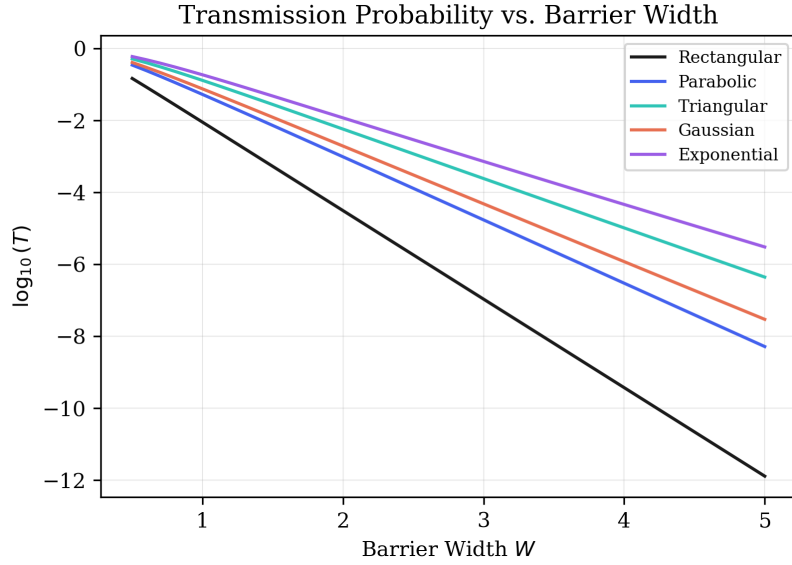


Figure 3: Logarithmic transmission probability $\log_{10}(T)$ versus barrier width W for the five geometries. Steeper slopes reflect larger integrated barrier areas.

3.3 The Shape Factor

The WKB approximation gives the tunneling probability as $T \sim \exp(-2\gamma)$, where

$$\gamma = \int_{\text{barrier}} \sqrt{2[V(x) - E]} dx. \quad (2)$$

We define the **Shape Factor** as the ratio of this integral for a given profile to the rectangular reference:

$$S = \frac{\gamma_{\text{shape}}}{\gamma_{\text{rect}}}. \quad (3)$$

Since $S < 1$ for any profile that dips below V_0 within its support, S directly measures how much “easier” a barrier is to tunnel through compared to the rectangle.

Table 4 compares the WKB-predicted Shape Factor with the actual numerical transmission at $W = 3.0$, and Figure 4 shows the empirical relationship between S and $\log_{10}(T)$.

Table 4: Shape Factor S and numerical transmission at $W = 3.0$, $V_0 = 5.0$, $E = 1.0$. The WKB integral correctly ranks the barriers in order of increasing transmission.

Barrier	Shape Factor S	WKB Integral γ	Numerical T
Rectangular	1.000	8.496	1.07×10^{-7}
Parabolic	0.702	5.962	1.71×10^{-5}
Gaussian	0.628	5.338	4.77×10^{-5}
Triangular	0.533	4.527	2.41×10^{-4}
Exponential	0.446	3.788	7.20×10^{-4}

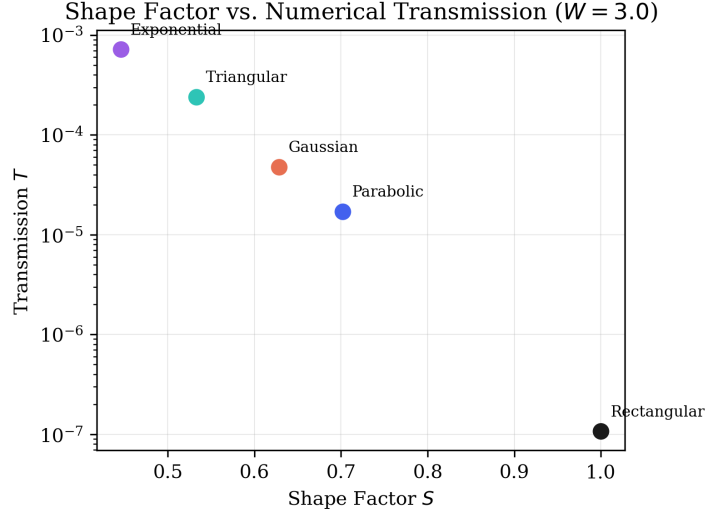


Figure 4: Correlation between Shape Factor S and numerical transmission probability T on a logarithmic scale at $W = 3.0$, demonstrating strong monotonic scaling.

As expected, S decreases monotonically and the numerical T increases monotonically in the same order. This confirms that the WKB integral captures the essential physics: a smaller effective barrier area means exponentially more tunneling.

Limitations of the WKB approach. The WKB approximation breaks down near classical turning points where $V(x) \approx E$. For smooth barriers (Gaussian, parabolic), this breakdown is mild because the wavefunction transitions gradually. For the rectangular barrier, the potential jumps discontinuously and the WKB connection formulas are needed. In practice, the WKB prediction provides the correct qualitative ranking and approximate magnitude, but the exact numerical ratio between barriers requires the full numerical calculation.

4 Additional Applications of the Solver

The following subsections demonstrate the versatility of the numerical framework. Each application uses the same eigenvalue solver described in Section 2, applied to a different potential.

4.1 Double-Well Tunnel Splitting

A symmetric double-well potential $V(x) = V_0(x^2 - a^2)^2/a^4$ was solved for the two lowest eigenstates. As shown in Figure 5, the ground state wavefunction is symmetric, and the first excited state is antisymmetric. At close well

separation ($a = 1.5$, $V_0 = 8.0$), the energy splitting is $\Delta E = E_1 - E_0 \approx 0.125$. As a increases and the central barrier grows, tunneling between the wells is suppressed and the splitting shrinks exponentially (reaching $\Delta E \approx 10^{-5}$ at large separations), consistent with the standard textbook result.

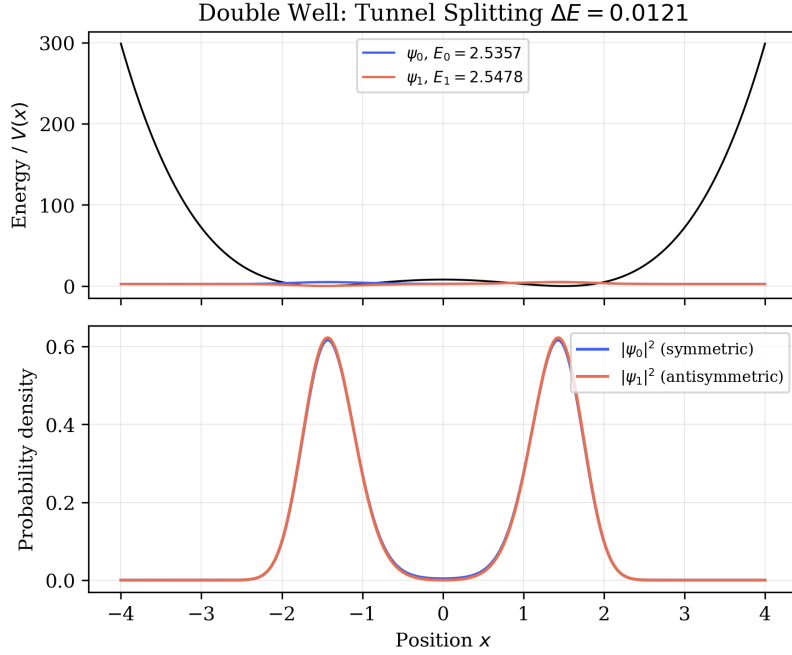


Figure 5: Double-well potential and the symmetric ground state (ψ_0) and antisymmetric first excited state (ψ_1). Bottom panel: corresponding probability densities.

4.2 Morse Potential and Molecular Vibrations

The Morse potential $V(x) = D[1 - e^{-a(x-x_0)}]^2$ models anharmonic molecular bonds. With $D = 10.0$, $a = 1.0$, $x_0 = 0$, $N = 500$ points on $x \in [-3, 8]$, Table 5 compares numerical and analytical eigenvalues, and Figure 6 plots the bound state wavefunctions offset by their energies.

Table 5: Morse potential energy levels. Analytical formula: $E_n = \omega(n + \frac{1}{2}) - \omega^2(n + \frac{1}{2})^2/(4D)$, where $\omega = a\sqrt{2D} = 4.472$. The first four bound states agree to better than 0.02%. States $n \geq 4$ are near or above the dissociation threshold $D = 10.0$.

State	Numerical	Analytical	ΔE	Error (%)
$n = 0$	2.111	2.111	—	0.012
$n = 1$	5.582	5.583	3.472	0.014
$n = 2$	8.054	8.055	2.472	0.014
$n = 3$	9.527	9.528	1.473	0.007

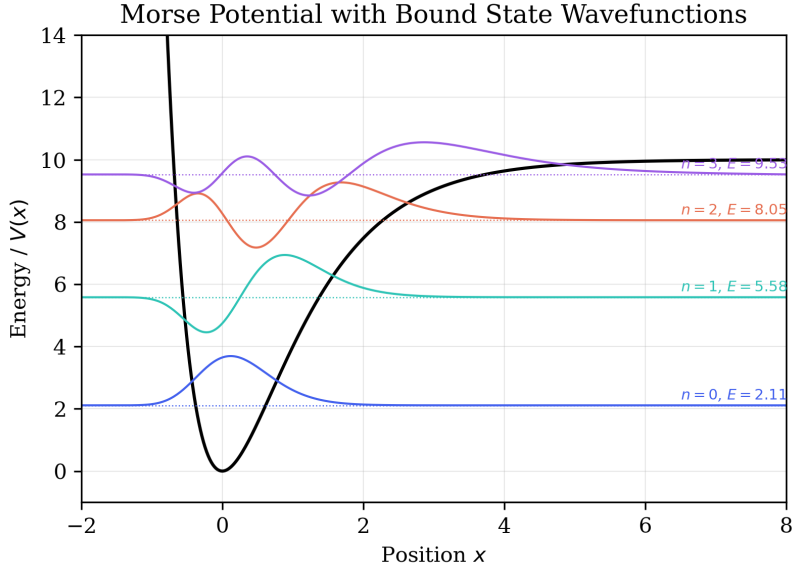


Figure 6: Morse potential with the lowest four calculated bound-state wavefunctions shifted to their respective energy eigenvalues. Energy level spacing ΔE visibly contracts.

The decreasing energy gap ($\Delta E = 3.47 \rightarrow 2.47 \rightarrow 1.47$) is the signature of anharmonicity. In a harmonic oscillator the gaps would all be equal. This compression of energy levels toward the dissociation energy D is what allows real molecules to dissociate when given enough vibrational energy.

4.3 Periodic Lattice and Band Structure

Replacing the single well with a periodic array of five attractive Gaussian potentials (depth 12.0, spacing 3.0, width $\sigma = 0.5$, on a grid of $N = 800$ points), the eigenvalue spectrum organizes into bands. The lowest five eigenvalues cluster tightly around $E \approx -11.5$ (the valence band), while the next

five span $E \in [-6.2, -5.0]$ (the conduction band). The gap between bands — the band gap — emerges naturally from the periodicity, illustrating how microscopic quantum mechanics gives rise to the electronic structure that determines whether a material is a conductor, semiconductor, or insulator.

4.4 Anderson Localization

Adding random disorder to the well depths (drawn uniformly from [8, 16] instead of a fixed 12.0) destroys the extended Bloch-like states. The wavefunctions become exponentially localized around individual wells, with the ground state trapped in the single deepest potential. This is consistent with Anderson localization in one dimension, where any amount of disorder is sufficient to localize all states.

4.5 Resonant Tunneling

Using the transfer matrix method on a double-barrier system (two rectangular barriers of height 5.0, width 0.5, separated by a gap of 2.0), we swept the incident energy E from 0.1 to 5.0 in steps of 0.005. As shown in Figure 7, at most energies $T < 10^{-3}$. However, sharp resonance peaks emerge where the transmission approaches unity ($T \rightarrow 1$). This occurs because the de Broglie wavelength of the particle matches the cavity length, producing constructive interference — the quantum analogue of a Fabry–Pérot resonator.

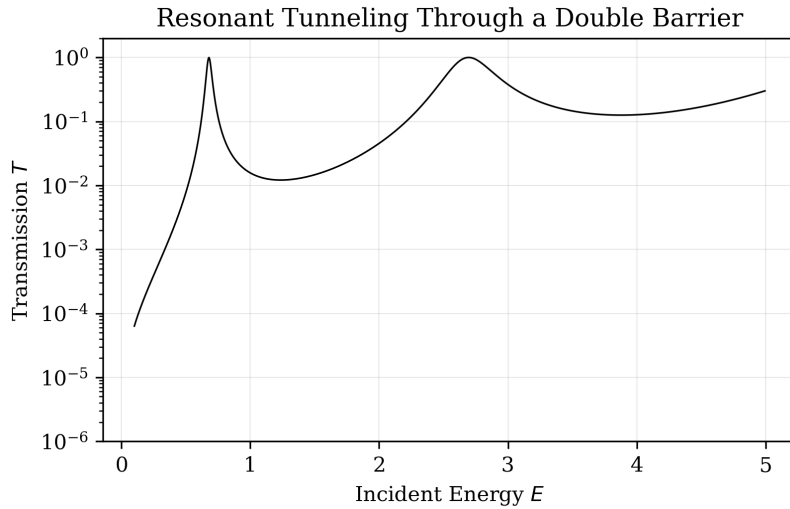


Figure 7: Transmission spectrum $T(E)$ across a double-barrier potential. Resonant transmission peaks approach unity where constructive interference occurs within the quantum cavity.

4.6 Wave Packet Scattering

The time-dependent Schrödinger equation was integrated using the Crank–Nicolson scheme, which is unconditionally stable and unitary (preserving the norm of the wavefunction to machine precision). Table 6 lists the simulation parameters, and Figure 8 displays sequential time snapshots.

Table 6: Wave packet simulation parameters.

Parameter	Value
Grid points N	400
Spatial domain	$x \in [-15, 15]$, $\Delta x = 0.075$
Time step Δt	0.01
Propagation steps	200 ($t_{\max} = 2.0$)
Initial position x_0	-5.0
Initial momentum k_0	5.0
Packet width σ	1.0
Barrier	Rectangular, $V_0 = 3.0$, width 1.0
Norm conservation	$< 10^{-6}$ drift

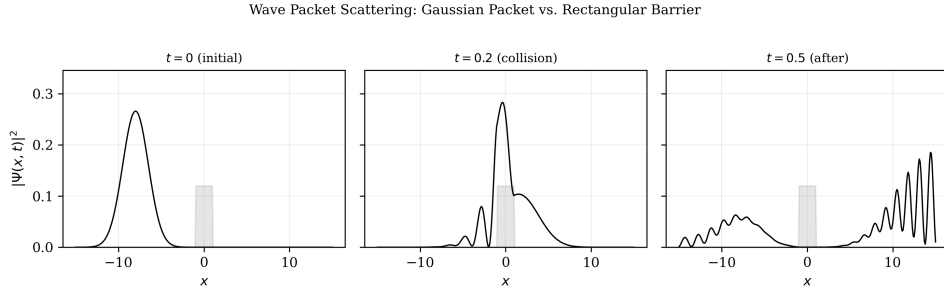


Figure 8: Temporal snapshots of probability density $|\Psi(x, t)|^2$ during wave packet collision with a rectangular barrier, illustrating spatial bifurcation into transmitted and reflected wave packets.

The key observation is that the wave packet does not behave classically. Instead of being entirely reflected or entirely transmitted, the probability density splits continuously into two components propagating in opposite directions. The norm is conserved to better than 10^{-6} over the entire simulation, confirming the numerical stability of the Crank–Nicolson integrator.

5 Discussion

The central result of this study is that barrier geometry introduces order-of-magnitude changes in tunneling probability, even when the peak height and width are held fixed. This is not a subtle correction. At $W = 5.0$ with $V_0 = 5.0$ and $E = 1.0$, the exponential barrier transmits over two million times more than the rectangular barrier. For any application where accurate tunneling rates matter — field emission, scanning tunneling microscopy, nuclear fusion, molecular reaction rates — the shape of the barrier cannot be ignored.

The Shape Factor S provides a useful rule of thumb. Since $\log T \propto -2\gamma$ in the WKB limit, and $\gamma_{\text{shape}} = S \cdot \gamma_{\text{rect}}$, the transmission through a non-rectangular barrier is approximately

$$T_{\text{shape}} \approx T_{\text{rect}}^S. \quad (4)$$

This power-law scaling is a simple and practical way to estimate tunneling corrections without re-solving the Schrödinger equation from scratch.

A few caveats are worth noting. First, all calculations were performed in natural units ($\hbar = m = 1$). Translating these results to physical systems (e.g., electron tunneling through a 1 nm oxide layer) requires restoring the appropriate mass and length scales. Second, the enormous 10^6 ratio we observe is specific to our chosen parameters ($V_0/E = 5$, $W = 5.0$); different ratios of barrier height to particle energy will produce different — though qualitatively similar — enhancement factors. Third, the WKB-based Shape Factor neglects interference effects and fails near classical turning points, which means it should be treated as an approximate guide rather than an exact predictor.

6 Conclusion

We developed a lightweight, dependency-free Schrödinger equation solver, verified it against analytical benchmarks with sub-0.04% error, and used it to systematically investigate the role of barrier geometry in quantum tunneling. Our main findings are:

1. Among five non-rectangular profiles tested, all produced substantially higher transmission than the rectangular barrier, with the exponential profile showing the largest enhancement (up to $\sim 10^6 \times$ at $W = 5.0$).
2. The WKB-derived Shape Factor S correctly ranks the barriers by transmission and provides a quantitative link between barrier area and tunneling rate.

3. The same numerical framework successfully reproduces tunnel splitting, Morse anharmonicity, band gaps, Anderson localization, resonant tunneling, and wave packet dynamics, demonstrating its generality.

The source code for the solver, all simulation scripts, and the raw data files are available at <https://github.com/dakshesh123-rgb/qm-project-website>.

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